

My own cloud for large scale data science

Cloud infrastructures provide various types of services, among them, resources for parallel and distributed computing and storage. In this assignment, your job is to build an infrastructure that can provide the following services:

- User registration, secure login and account management
- Provision of computational resources that is elastic (shrinks and grows as needed)
- Disk storage
- Container service
- Database service

Your implementation should implement some reasonable User Level Agreement, so that jobs submitted by the user can run in a reasonable time with a near optimal number of devices. It would also be helpful if the implementation saves energy as much as possible.

You can deploy your cloud services using Eucalyptus, Open Nebula, OpenStack etc or can implement your own services. If you use Eucalyptus, it is recommended to download a

`faststart`

version to inspect and test before performing a full installation (more details can be found [here](#)). If you use Open Nebula, you can start [here](#).

Please, not that these software already provide services, but your job is to re-implement them with your own code and explain how the services were implemented.

Suggested structure for the report:

1. Brief background on the chosen software and libraries
2. Materials and methods
 1. Machines used and their characteristics
 2. Software used and their versions
 3. Detailed guide of how the cloud was developed
 1. Installation steps
 2. Selected tests
 3. Evaluation/assessment metrics
3. Discussion and conclusions

Deadline: June 3rd, 2025